

manufacture of small cameras, according to the researchers.

The result is a compound lens just 0.1-millimetre wide, which only took a few hours to design, manufacture and test. It can focus on images from a distance of three millimetres, and relay these down the length of a 1.7-metre optical fibre that it is attached to.

The whole system fits comfortably into the needle of a standard syringe, allowing it to be injected into the human body, or even the brain. From there it can act as an endoscope, giving doctors the ability to examine the inside of the body directly.

Bacteria-powered batteries for smartphones

A team of scientists from Oxford University, UK, has shown how the natural movement of bacteria could be harnessed to assemble and power microscopic windfarms or other man-made micro-machines such as smartphone components.

The study used computer simulations to demonstrate that the chaotic swarming effect of dense active matter such as bacteria can be organised to turn cylindrical rotors and provide a steady power source.

These biologically-driven power plants could someday become microscopic engines for tiny, man-made devices that are self-assembled and self-powered—everything from optical switches to smartphone microphones.

Octopus-like smart adhesive sensors

A team of researchers from South Korea has developed a flexible-pressure sensor by mimicking the suction cups on an octopus's tentacles. Researchers from Ulsan National Institute

of Science And Technology (UNIST) and Korea Institute of Science and Technology (KIST) studied the adhesive mechanism of octopus suckers and used it to develop a new type of suction based adhesive material.

By mimicking muscle actuation to control cavity-pressure-induced adhesion of octopus



The scientists hope that their product can be used as the substrate for wearable health sensors or band-aids that stick to the skin at normal body temperatures but fall off when rinsed under cold water (Image courtesy: www.vishwagujarat.com)

suckers, Hyunhyub Ko from UNIST has used a unique rubbery material to create an array of micro-scale suckers, which include pores that are coated with a thermally-responsive polymer to create sucker-like walls.

Engineered sand to cool electronic devices

A USA based researcher has devised a new type of sand that can provide improved cooling for power-hungry electronic devices. Baratunde Cola, associate professor at Georgia Institute of Technology, USA, has devised silicon-dioxide nanoparticles coated with a high-dielectric-constant polymer that could be used potentially for heat dissipation from power electronics, LEDs and other applications with high heat fluxes.

According to the findings, the silicon-dioxide does not do the cooling itself and the unique surface properties of the coated nano-scale material conducts heat at potentially higher efficiency than existing heat-sink materials. Using the collective surface electromagnetic effect of the nanoparticles, thermal conductivity can increase 20-fold, allowing it to dissipate heat much faster.

The researchers have used ethylene glycol, a fluid commonly used in vehicle antifreeze, as a heat-dissipation material that, at the same time, does not conduct electricity.

Humans, robots work in harmony at Ford Cologne factory

Most factory robots do not have the ability to stop or subtly adjust their motion, so anyone who gets in the way is at risk of injury. Ford has found a way around this problem, and is putting humans side-by-side with robots in its Cologne factory to validate its approach.



Ford assembly-line staff in Europe is working hand-in-hand with collaborative robots, known as co-bots, to fit shock absorbers to Fiesta cars (Image courtesy: <https://media.ford.com>)

Ford's new robots are being used to fit shock absorbers to Fiestas. Rather than forcing an assembly-line worker to manipulate a heavy shock absorber above their head, workers can use the robots to lift and position the part, before finishing the operation with the push of a button. The robots have been programmed to be conservative when it comes to human contact. If an arm or finger gets in the way, sensors detect the obstruction and the machine stops moving to protect the worker.